Enterprise Network Lab

Multi-Floor Office Network with Routing, Switching & Firewall Integration

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Portfolio Project Network Security Engineering Track

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1. Project Overview

This document presents a full enterprise network lab built and verified in EVE-NG, a professional network emulation platform running on VMware Workstation. The lab simulates a real multi-floor office environment with proper network segmentation, dynamic routing, centralized DHCP, and perimeter firewall security.

The project demonstrates hands-on competency across the full enterprise network stack from physical cabling and Layer 2 switching at the access layer, through Layer 3 inter-VLAN routing at the core, to firewall policy configuration and internet NAT at the perimeter. Every configuration decision reflects real-world engineering practice rather than simplified lab shortcuts.

1.1 Objectives

- Design and implement a multi-VLAN enterprise network with floor-based segmentation
- Configure a hierarchical switching topology using VTP for centralized VLAN management
- Implement Inter-VLAN routing using a Layer 3 core switch with SVIs
- Deploy OSPF dynamic routing between the core switch and edge router
- Configure centralized DHCP on the router with relay agents on the core switch
- Integrate a FortiGate next-generation firewall for perimeter security and NAT
- Verify full end-to-end connectivity from end hosts through to the internet

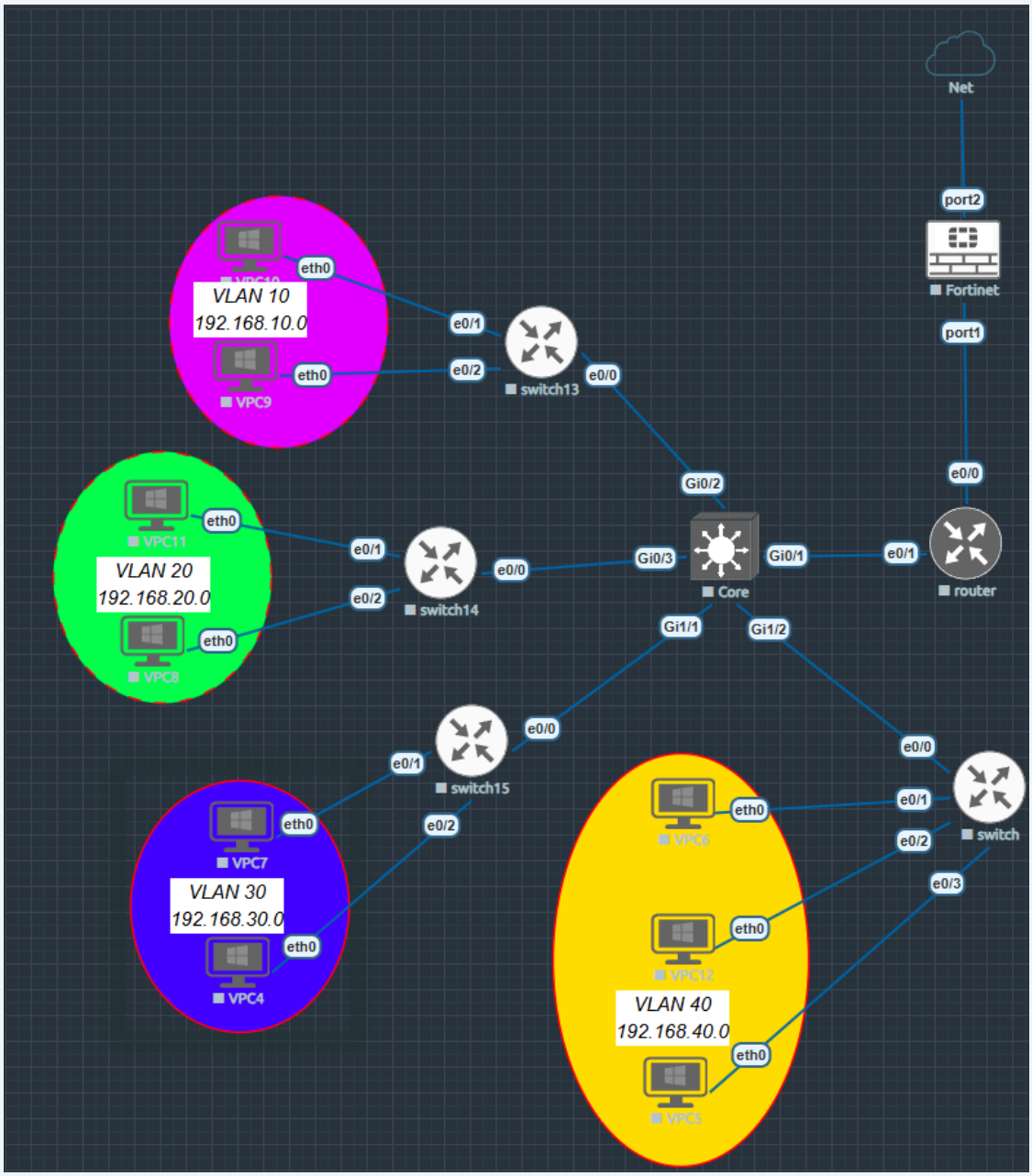
1.2 Platform & Environment

Component	Details
Emulation Platform	EVE-NG Community Edition
Host Environment	VMware Workstation (VMnet8 NAT)
EVE-NG Host IP	192.168.202.141
Internet Gateway	VMware NAT, 192.168.202.1
Operating System	Windows 11 Host

2. Network Topology

The topology follows a three-tier hierarchical design, a standard approach used in enterprise networks to ensure scalability, fault isolation, and clear traffic flow:

- Access Layer: Four IOU Layer 2 switches connecting end-host VPCs, one per floor
- Core/Distribution Layer: One vIOS-L2 Layer 3 switch providing inter-VLAN routing
- Edge Layer: One IOU router connecting to the perimeter firewall
- Perimeter: FortiGate VM64 firewall separating the internal network from the internet



2.1 Logical Topology Diagram

```
Internet (VMware NAT Cloud)
|
[ FortiGate VM64 ]
port2 (WAN) --- port1 (LAN)
192.168.202.x      192.168.200.2/30
|
[ Edge Router ]
e0/0: 192.168.200.1/30
e0/1: 192.168.100.2/30
|
[ Core Switch (L3) ]
Gi0/1: 192.168.100.1/30
Gi0/2    Gi0/3    Gi1/1    Gi1/2
|        |        |        |
SW13     SW14     SW15     SW(Yellow)
VLAN10   VLAN20   VLAN30   VLAN40
|        |        |        |
VPC9,10  VPC8,11  VPC3,7   VPC5,6,12
```

2.2 Device Inventory

Device	Role	Image	Type
Core Switch	L3 Core / Distribution	viosl2-adventerprisek9 high_iron_20200929	QEMU
SW13, SW14, SW15, SW(Yellow)	Access Layer	i86bi_linux_l2 high_iron_20180510	IOU
Router	Edge Router / DHCP Server	i86bi_LinuxL3-AdvEnterpriseK9 157_3_May_2018	IOU
FortiGate	Perimeter Firewall / NAT	FortiGate-VM64-KVM	QEMU
VPCs (x12)	End Hosts	EVE-NG Built-in VPCS	—

3. IP Addressing Scheme

A structured IP addressing scheme was designed before implementation. Point-to-point infrastructure links use /30 subnets (4 IPs, 2 usable) to avoid wastage. Floor VLANs use /24 subnets to accommodate up to 254 hosts per floor.

Segment	Network	Gateway / IP	Notes
VLAN 10 Finance (Pink)	192.168.10.0/24	192.168.10.1	Core SVI SW13 VPCs
VLAN 20 HR (Green)	192.168.20.0/24	192.168.20.1	Core SVI SW14 VPCs
VLAN 30 Marketing (Blue)	192.168.30.0/24	192.168.30.1	Core SVI SW15 VPCs
VLAN 40 IT (Yellow)	192.168.40.0/24	192.168.40.1	Core SVI SW(Yellow) VPCs
Core Uplink (Core ↔ Router)	192.168.100.0/30	.1 Core / .2 Router	Routed port Gi0/1
Router ↔ FortiGate Link	192.168.200.0/30	.1 Router / .2 FortiGate	e0/0 ↔ port1
FortiGate WAN (port2)	192.168.202.0/24	192.168.202.1 (NAT GW)	DHCP from VMware NAT

Note: /30 subnets are used on point-to-point infrastructure links there are only ever two devices on each link, so a /30 (2 usable IPs) is the most efficient choice. Using a /24 on a two-device link would waste 252 addresses.

4. Layer 2 Configuration

4.1 VLAN Design

Four VLANs were created to segment network traffic by floor/department. This ensures that broadcast traffic from one floor does not flood other floors, improves security by isolating departments, and simplifies network management.

VLAN ID	Name	Floor	Access Switch
VLAN 10	FINANCE	Pink Floor	Switch13
VLAN 20	HR	Green Floor	Switch14
VLAN 30	MARKETING	Blue Floor	Switch15
VLAN 40	IT	Yellow Floor	Switch (Yellow)

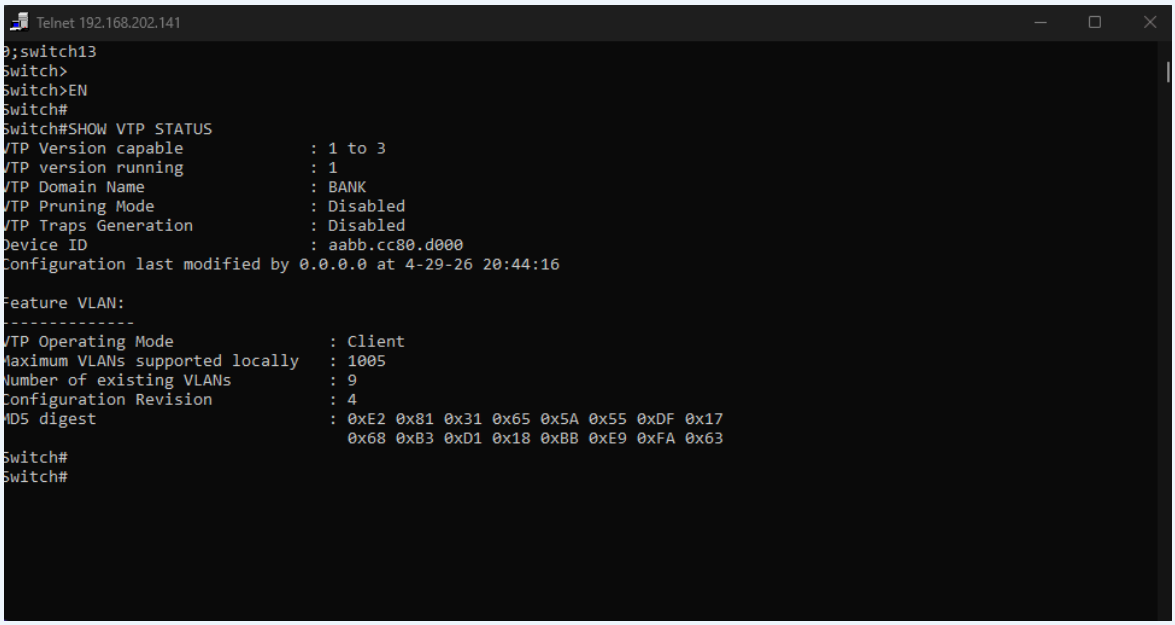
```
Switch>
Switch>EN
Switch#
Switch#SHOW VLAN BR

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gi0/0, Gi1/0, Gi1/3, Gi2/0
10   FINANCE                          active
20   HR                               active
30   MARKETING                        active
40   IT                               active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
Switch#
Switch#
Switch#
Switch#
```

4.2 VTP (VLAN Trunking Protocol)

VTP was implemented to centralize VLAN management. Instead of manually creating VLANs on each of the four access switches, VTP automatically propagates the VLAN database from the server (Core switch) to all client switches across the domain.

- Core Switch VTP Mode: Server | Domain: BANK
- All Access Switches VTP Mode: Client | Domain: BANK
- VLANs are propagated automatically no manual VLAN creation on access switches

Security Note: VTP only operates over trunk links. The fact that VLANs were visible on access switches before trunk ports were explicitly configured revealed that DTP (Dynamic Trunking Protocol) had auto-negotiated trunks. In production environments, DTP is disabled using switchport nonegotiate to prevent unauthorized switches from forming trunks and receiving VLAN data a VLAN hopping attack vector.



4.3 Trunk Port Configuration

All uplink ports between the Core switch and access switches were explicitly configured as 802.1Q trunks. Explicit configuration was chosen over DTP auto-negotiation for security and predictability.

Core Switch Trunk Ports

```
interface range GigabitEthernet0/2 - 3, GigabitEthernet1/1 - 2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport nonegotiate
```

Access Switch Uplink (e0/0 on each access switch)

```
interface Ethernet0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport nonegotiate
```

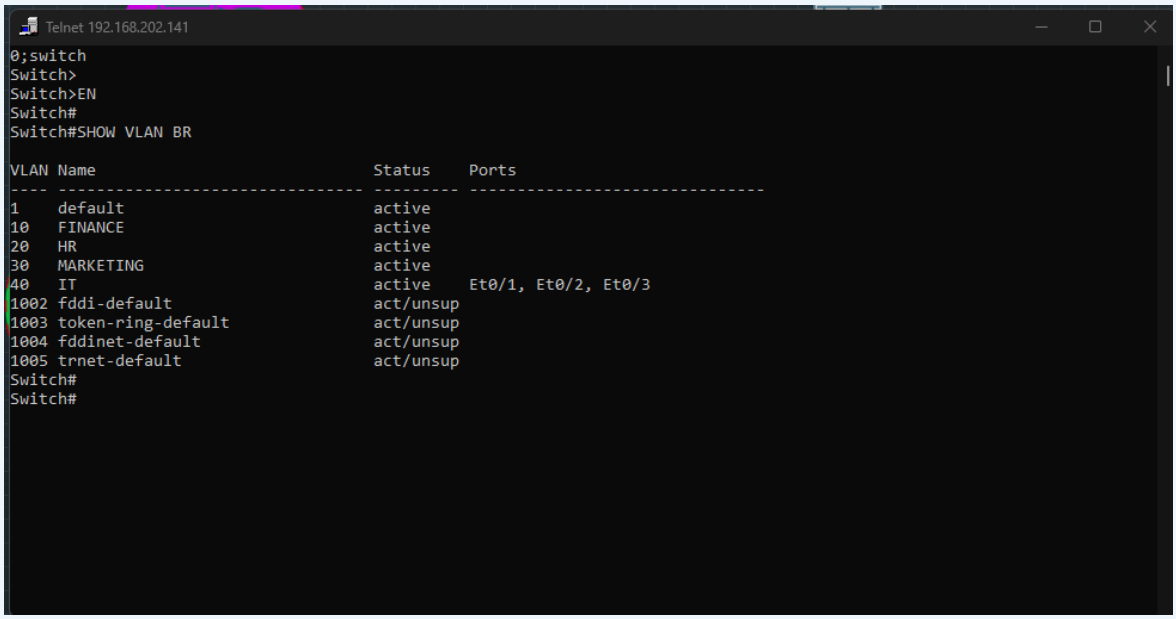
```
Telnet 192.168.202.141
0;Core
Switch#
Switch#EN
Switch#
Switch#SHOW INTERFACE TRUNK

Port        Mode             Encapsulation  Status        Native vlan
Gi0/2       on               802.1q         trunking      1
Gi0/3       on               802.1q         trunking      1
Gi1/1       on               802.1q         trunking      1
Gi1/2       on               802.1q         trunking      1

Port        Vlans allowed on trunk
Gi0/2       1-4094
Gi0/3       1-4094
Gi1/1       1-4094
Gi1/2       1-4094

Port        Vlans allowed and active in management domain
Gi0/2       1,10,20,30,40
Gi0/3       1,10,20,30,40
Gi1/1       1,10,20,30,40
Gi1/2       1,10,20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Gi0/2       1,10,20,30,40
Gi0/3       1,10,20,30,40
Gi1/1       1,10,20,30,40

Switch#
```

4.4 Access Port Configuration

VPC-facing ports on each access switch were explicitly configured as access ports and assigned to the correct VLAN. switchport nonegotiate was applied to prevent DTP negotiation on end-host ports.

Example Switch13 (VLAN 10, Finance Floor)

```
interface range Ethernet0/1 - 2
 switchport mode access
 switchport access vlan 10
 switchport nonegotiate
```



5. Layer 3 Configuration

5.1 Core Switch Layer 3 Routing

The Core switch (vIOS-L2) was configured as a Layer 3 switch to perform inter-VLAN routing. This is more efficient than using a separate router for inter-VLAN traffic because switching happens in hardware at wire speed.

ip routing was enabled globally to activate the Layer 3 routing engine:

```
ip routing
```

SVIs (Switched Virtual Interfaces)

One SVI was created per VLAN, each acting as the default gateway for that floor's VPCs:

```
interface Vlan10
 ip address 192.168.10.1 255.255.255.0
 ip helper-address 192.168.100.2
 no shutdown
!
interface Vlan20
 ip address 192.168.20.1 255.255.255.0
 ip helper-address 192.168.100.2
 no shutdown
!
interface Vlan30
 ip address 192.168.30.1 255.255.255.0
```

```
 ip helper-address 192.168.100.2
 no shutdown
!
interface Vlan40
 ip address 192.168.40.1 255.255.255.0
 ip helper-address 192.168.100.2
 no shutdown
```

Routed Uplink Port to Router

Gi0/1 was converted from a switchport to a routed port using no switchport, then assigned an IP for the point-to-point link to the router:

```
interface GigabitEthernet0/1
 no switchport
 ip address 192.168.100.1 255.255.255.252
 no shutdown
```

```
Switch#
Switch#
Switch#SHOW IP INTERFACE BRIEF
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/1     192.168.100.1   YES NVRAM  up                    up
GigabitEthernet0/2     unassigned      YES unset  up                    up
GigabitEthernet0/3     unassigned      YES unset  up                    up
GigabitEthernet1/0     unassigned      YES unset  up                    up
GigabitEthernet1/1     unassigned      YES unset  up                    up
GigabitEthernet1/2     unassigned      YES unset  up                    up
GigabitEthernet1/3     unassigned      YES unset  up                    up
GigabitEthernet2/0     unassigned      YES unset  up                    up
Vlan10                 192.168.10.1    YES NVRAM  up                    up
Vlan20                 192.168.20.1    YES NVRAM  up                    up
Vlan30                 192.168.30.1    YES NVRAM  up                    up
Vlan40                 192.168.40.1    YES NVRAM  up                    up
Switch#
Switch#
```

5.2 Router Configuration

Interface IP Assignment

```
interface Ethernet0/1
 ip address 192.168.100.2 255.255.255.252
 no shutdown
!
interface Ethernet0/0
 ip address 192.168.200.1 255.255.255.252
 no shutdown
```

5.3 OSPF Dynamic Routing

OSPF (Open Shortest Path First) was configured between the Core switch and the router to dynamically exchange routing information. This means that if new subnets are added in the future, they will be automatically advertised without manual static route updates.

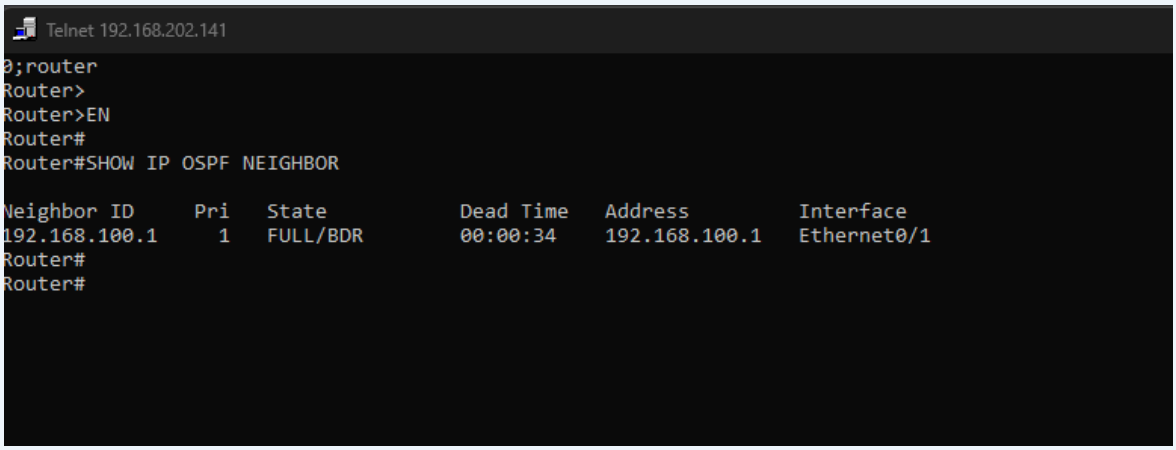
Core Switch OSPF Configuration

```
router ospf 1
 network 192.168.10.0 0.0.0.255 area 0
 network 192.168.20.0 0.0.0.255 area 0
 network 192.168.30.0 0.0.0.255 area 0
 network 192.168.40.0 0.0.0.255 area 0
 network 192.168.100.0 0.0.0.3 area 0
```

Router OSPF Configuration

```
router ospf 1
 network 192.168.100.0 0.0.0.3 area 0
 network 192.168.200.0 0.0.0.3 area 0
 default-information originate
```

The default-information originate command redistributes the router's default static route into OSPF, allowing the Core switch to automatically learn the path to the internet without requiring a manually configured static route on the Core.



5.4 Routing Tables

Router Routing Table

The following output was captured from the router after full configuration. It shows OSPF-learned routes for all four VLAN subnets and the static default route toward the FortiGate:

```
Router#SHOW IP ROUTE
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 192.168.200.2 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 192.168.200.2
O     192.168.10.0/24 [110/11] via 192.168.100.1, 00:13:20, Ethernet0/1
O     192.168.20.0/24 [110/11] via 192.168.100.1, 00:13:20, Ethernet0/1
O     192.168.30.0/24 [110/11] via 192.168.100.1, 00:13:20, Ethernet0/1
O     192.168.40.0/24 [110/11] via 192.168.100.1, 00:13:20, Ethernet0/1
      192.168.100.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.100.0/30 is directly connected, Ethernet0/1
L        192.168.100.2/32 is directly connected, Ethernet0/1
      192.168.200.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.200.0/30 is directly connected, Ethernet0/0

Router#
```

Key routes visible in the router routing table:

- S* 0.0.0.0/0 [1/0] via 192.168.200.2 Default route toward FortiGate (internet)
- O 192.168.10.0/24 via 192.168.100.1 VLAN 10 learned via OSPF from Core
- O 192.168.20.0/24 via 192.168.100.1 VLAN 20 learned via OSPF from Core
- O 192.168.30.0/24 via 192.168.100.1 VLAN 30 learned via OSPF from Core
- O 192.168.40.0/24 via 192.168.100.1 VLAN 40 learned via OSPF from Core
- C 192.168.100.0/30 Directly connected Core-Router link
- C 192.168.200.0/30 Directly connected Router-FortiGate link

Core Switch Routing Table

The Core switch routing table shows all four VLAN subnets as directly connected routes, plus the OSPF-learned default route from the router:

```
Switch#SHOW IP ROUTE
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 192.168.100.2 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/10] via 192.168.100.2, 00:11:17, GigabitEthernet0/1
      192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.10.0/24 is directly connected, Vlan10
L        192.168.10.1/32 is directly connected, Vlan10
      192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.20.0/24 is directly connected, Vlan20
L        192.168.20.1/32 is directly connected, Vlan20
      192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.30.0/24 is directly connected, Vlan30
L        192.168.30.1/32 is directly connected, Vlan30

Switch#
```

- C 192.168.10.0/24 via Vlan10 VLAN 10 Finance floor
- C 192.168.20.0/24 via Vlan20 VLAN 20 HR floor
- C 192.168.30.0/24 via Vlan30 VLAN 30 Marketing floor
- C 192.168.40.0/24 via Vlan40 VLAN 40 IT floor
- C 192.168.100.0/30 via Gi0/1 Routed uplink to router
- O*E2 0.0.0.0/0 via 192.168.100.2 Default route learned via OSPF from router

6. DHCP Configuration

6.1 DHCP Relay (ip helper-address)

DHCP uses broadcast packets. Broadcasts do not cross Layer 3 boundaries meaning a DHCP discover from a VPC on VLAN 10 would never reach the DHCP server on the router without assistance.

The ip helper-address command was configured on each Core switch SVI to solve this. When the SVI receives a DHCP broadcast, it converts it to a unicast packet and forwards it to the router's IP address (192.168.100.2), which hosts the DHCP server.

```
Router#
Router#
Router#
Router#
Router#show running-config
Building configuration...

Current configuration : 1700 bytes
!
version 15.7
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname Router
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
mmi polling-interval 60
no mmi auto-configure
no mmi pvc
mmi snmp-timeout 180
```

6.2 DHCP Pools on Router

Four DHCP pools were configured on the router, one per VLAN. Each pool specifies the network, default gateway, DNS server, and excluded addresses:

```
ip dhcp excluded-address 192.168.10.1
ip dhcp excluded-address 192.168.20.1
ip dhcp excluded-address 192.168.30.1
ip dhcp excluded-address 192.168.40.1
!
ip dhcp pool VLAN10
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8
!
```

```
ip dhcp pool VLAN20
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 8.8.8.8
!
ip dhcp pool VLAN30
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 8.8.8.8
!
ip dhcp pool VLAN40
 network 192.168.40.0 255.255.255.0
 default-router 192.168.40.1
 dns-server 8.8.8.8
```

```
Router#show ip dhcp pool

Pool VLAN 10 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 192.168.10.3         192.168.10.1     - 192.168.10.254    1

Pool VLAN 20 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 192.168.20.1         192.168.20.1     - 192.168.20.254    0

Pool VLAN 30 :
 Utilization mark (high/low)    : 100 / 0

Router#
```

```
0;VPC6
Welcome to Virtual PC Simulator, version 1.3 (0.8.1)
Dedicated to Daling.
Build time: Feb 22 2024 06:25:41
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
Copyright (c) 2021, Alain Degreffe (alain.degreffe@eve-ng.net)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version for EVE-NG.

Press '?' to get help.

VPCS>
VPCS> dhcp
DDORA IP 192.168.40.2/24 GW 192.168.40.1

VPCS>
```

7. FortiGate Firewall Configuration

7.1 Interface Configuration

The FortiGate separates the internal network from the internet. Two interfaces were configured:

- port1 (LAN/Inside) Static IP 192.168.200.2/30, role lan, allowaccess ping https ssh
- port2 (WAN/Outside) DHCP from VMware NAT cloud, role wan, allowaccess ping only

Setting port2 to allowaccess ping only (not https/ssh) is a security best practice management access should never be exposed to the WAN interface in production environments.

```
config system interface
  edit port1
    set mode static
    set ip 192.168.200.2 255.255.255.252
    set allowaccess ping https ssh
    set role lan
  next
  edit port2
    set mode dhcp
    set allowaccess ping
    set role wan
  next
end
```

```
FortiGate-VM64-KVM login: admin
Password:
Welcome!

FortiGate-VM64-KVM # get system interface physical
== [onboard]
        ==[port1]
                mode: static
                ip: 192.168.200.2 255.255.255.252
                ipv6: ::/0
                status: up
                speed: 10000Mbps (Duplex: full)
                FEC: none
                FEC_cap: none
        ==[port2]
                mode: dhcp
                ip: 192.168.202.143 255.255.255.0
                ipv6: ::/0
                status: up
                speed: 10000Mbps (Duplex: full)
                FEC: none
                FEC_cap: none
        ==[port3]
                mode: static
                ip: 0.0.0.0 0.0.0.0
                ipv6: ::/0
                status: up
                speed: 10000Mbps (Duplex: full)
                FEC: none
                FEC_cap: none
        ==[port4]
                mode: static
                ip: 0.0.0.0 0.0.0.0
                ipv6: ::/0
                status: up
                speed: 10000Mbps (Duplex: full)
                FEC: none
                FEC_cap: none

FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
```

7.2 Static Routing

Two static routes were configured on the FortiGate:

- Default route (0.0.0.0/0) out port2 toward VMware NAT gateway (192.168.202.1) for internet traffic
- Return route (192.168.0.0/16) via router (192.168.200.1) on port1 for traffic returning to internal VLANs

The 192.168.0.0/16 summary route covers all internal VLAN subnets (10.x, 20.x, 30.x, 40.x, 100.x, 200.x) in a single statement. This is route summarization, avoiding six separate static routes.

```
config router static
  edit 1
    set gateway 192.168.202.1
    set device port2
  next
  edit 2
    set dst 192.168.0.0 255.255.0.0
    set gateway 192.168.200.1
    set device port1
  next
end
```

```
FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM # show router static
config router static
    edit 1
        set gateway 192.168.202.1
        set device "port2"
    next
    edit 2
        set dst 192.168.0.0 255.255.0.0
        set gateway 192.168.200.1
        set device "port1"
    next
end
```

7.3 Firewall Policy LAN to WAN

FortiGate denies all traffic by default. An explicit allow policy was created to permit internal VLAN traffic to reach the internet:

```
config firewall policy
  edit 1
    set name "LAN-to-WAN"
    set srcintf "port1"
    set dstintf "port2"
    set srcaddr "all"
    set dstaddr "all"
    set action accept
    set schedule "always"
    set service "ALL"
    set nat enable
  next
end
```

NAT was enabled on this policy without it, private VPC IPs (192.168.10.x, etc.) would be sent directly to the internet, which would not route back since these are private RFC1918 addresses. NAT translates them to the FortiGate's public WAN IP before forwarding.

```
FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM # show firewall policy
config firewall policy
    edit 1
        set name "LAN-to-WAN"
        set uuid cfd24ed8-4567-51f1-089b-1c45a79ab8bd
        set srcintf "port1"
        set dstintf "port2"
        set action accept
        set srcaddr "all"
        set dstaddr "all"
        set schedule "always"
        set service "ALL"
        set nat enable
    next
end
```

8. Connectivity Verification

After completing all the configurations, systematic end-to-end testing was performed to verify every layer of the network stack.

8.1 VPC to Gateway (Layer 2 → Layer 3)

Each VPC was tested by pinging its default gateway (the Core switch SVI). A successful ping confirms the Layer 2 path from VPC through the access switch trunk to the Core is fully operational.

```
0;VPC7
VPCS> ping 192.168.30.1

84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=2.555 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=2.164 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=2.187 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=2.223 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=2.265 ms

VPCS>
```

8.2 Inter-VLAN Routing

A VPC on VLAN 10 was tested by pinging a VPC on VLAN 20. A successful ping confirms that the Core switch SVI routing is functioning correctly across VLAN boundaries.

```
0;VPC9
Welcome to Virtual PC Simulator, version 1.3 (0.8.1)
Dedicated to Daling.
Build time: Feb 22 2024 06:25:41
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
Copyright (c) 2021, Alain Degreffe (alain.degreffe@eve-ng.net)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version for EVE-NG.

Press '?' to get help.

VPCS>
VPCS> dhcp
DDORA IP 192.168.10.3/24 GW 192.168.10.1

VPCS> ping 192.168.20.2

84 bytes from 192.168.20.2 icmp_seq=1 ttl=63 time=2.358 ms
84 bytes from 192.168.20.2 icmp_seq=2 ttl=63 time=3.043 ms
84 bytes from 192.168.20.2 icmp_seq=3 ttl=63 time=2.767 ms
84 bytes from 192.168.20.2 icmp_seq=4 ttl=63 time=3.005 ms
84 bytes from 192.168.20.2 icmp_seq=5 ttl=63 time=2.516 ms

VPCS>
```

8.3 Full End-to-End Internet Connectivity

The ultimate verification test a VPC pinging 8.8.8.8 (Google DNS). This single test validates the entire network stack:

- VPC sends ICMP to 8.8.8.8
- Access switch forwards via trunk to Core
- Core switch routes via OSPF default route to Router
- Router forwards via static default route to FortiGate
- FortiGate applies NAT and LAN-to-WAN policy, forwards out port2
- VMware NAT forwards to the actual internet
- Reply returns through the same path in reverse

```
VPCS>
VPCS>
VPCS> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=125 time=73.428 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=125 time=75.161 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=125 time=70.791 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=125 time=72.053 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=125 time=65.223 ms

VPCS>
```

9. Troubleshooting Log

Several issues were encountered and systematically diagnosed during the lab. Each is documented below as a learning record and reference.

Issue	Root Cause	Resolution
SVIs administratively down	no shutdown not applied after SVI creation	Applied no shutdown to all four SVIs explicitly
No trunks forming on Core	Core downlink ports left at default switchport mode	Configured explicit trunk on Gi0/2, Gi0/3, Gi1/1, Gi1/2
DHCP failing can't find server	ip helper-address missing on SVIs	Added helper-address 192.168.100.2 on all four SVIs
Config lost after pause/restart	wr not run after configuration sessions	Established wr discipline after every config block
Core not routing internet traffic	Default route not propagated to Core switch	Added default-information originate on router OSPF

Issue	Root Cause	Resolution
VPCs back on VLAN 1 after restart	Access switch configs not saved with wr	Reconfigured and saved all access switches
FortiGate console not responding	Telnet console input handling issue	Switched to HTML5 console via EVE-NG web interface
ASAv console dead	ASAv 9.20 + EVE-NG compatibility issue	Replaced with FortiGate VM (more relevant for NSE path)

10. Skills Demonstrated

Skill Area	Specifics
Network Design	Hierarchical three-tier topology, IP addressing scheme, and subnet planning
Layer 2 Switching	VLAN creation, 802.1Q trunking, access port config, DTP hardening
VTP	Server/Client domain configuration, VLAN propagation verification
Spanning Tree	Root bridge election and STP verification
Layer 3 Switching	SVIs, routed ports, IP routing, inter-VLAN routing
OSPF	Single-area configuration, neighbor verification, and default route redistribution
DHCP	Pool configuration, exclusions, relay agent (ip helper-address)
FortiGate Firewall	Interface config, static routing, firewall policy, NAT
Troubleshooting	Systematic diagnosis using show commands, layer-by-layer verification
EVE-NG	Topology design, node management, console troubleshooting

About the Author

Timothy Lutara is a BSc Information Technology (Cybersecurity) student at ISBAT University in Kampala, Uganda, with a background in software engineering and a prior Diploma in Cybersecurity. He completed a cybersecurity internship at the Uganda Communications Commission (UCC) in 2024.

He is building toward a career as a Network Security Engineer with a certification roadmap targeting ISC2 CC, CompTIA CySA+, CCNA, and CCNP Security. This lab is part of a structured portfolio designed to demonstrate enterprise networking and security competency to prospective employers.

For more projects and documentation, visit his GitHub portfolio repository: nsec-portfolio